



# Interview Outcome Classification with BERT

Fine-tuning ruBERT-base to automatically  
classify recruitment call outcomes into  
five categories — a data analyst case study.

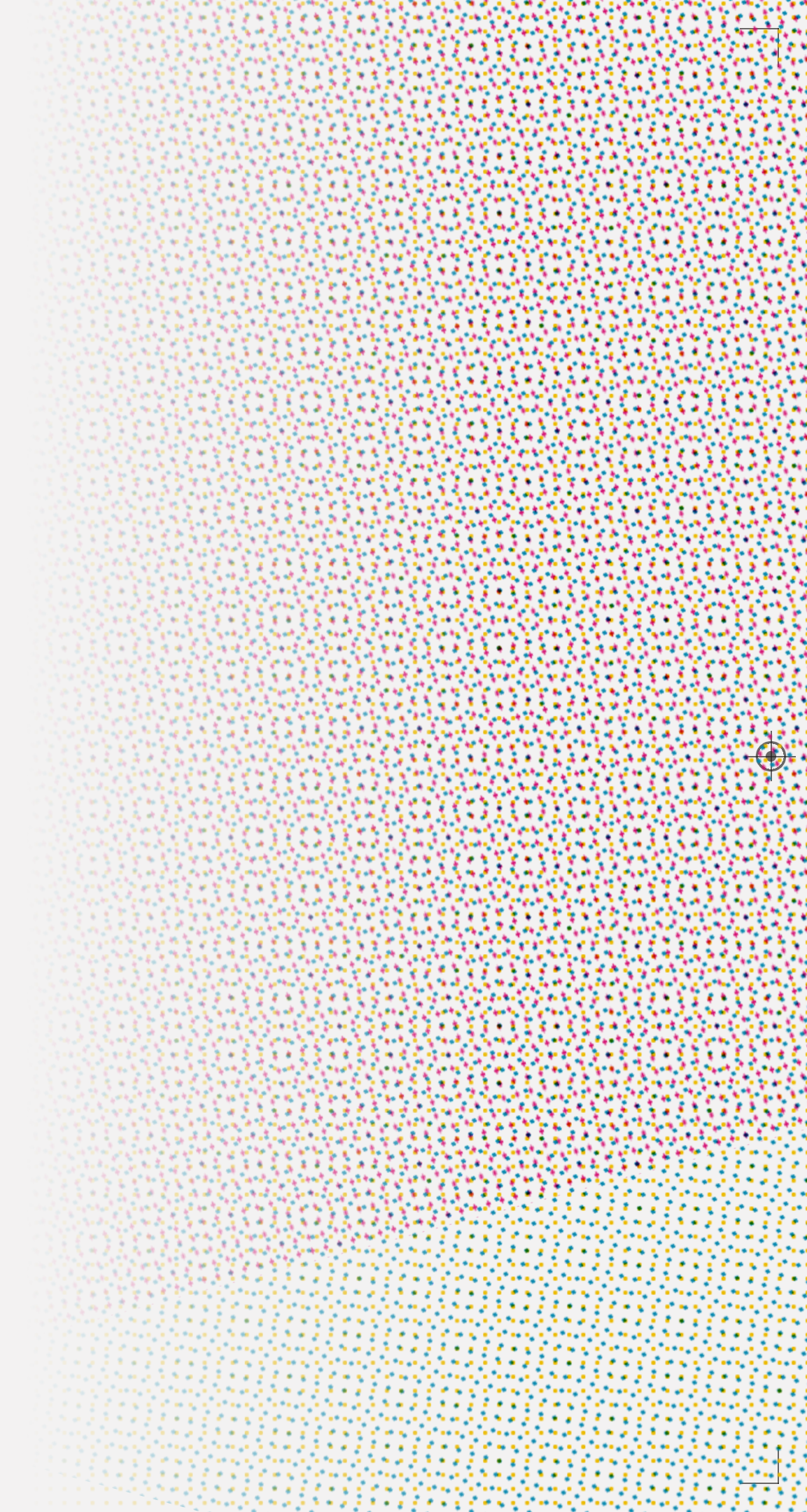


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# The problem

01



# Recruiter call outcomes are labelled by hand

- Recruitment teams conduct hundreds of phone interviews a week, and every call outcome is logged manually.
- Manual labelling is slow and inconsistent — the same outcome gets tagged differently across recruiters.
- Goal: train a Russian-language model to classify a call transcript into a structured outcome, enabling downstream analytics, reporting and recruiter performance tracking.

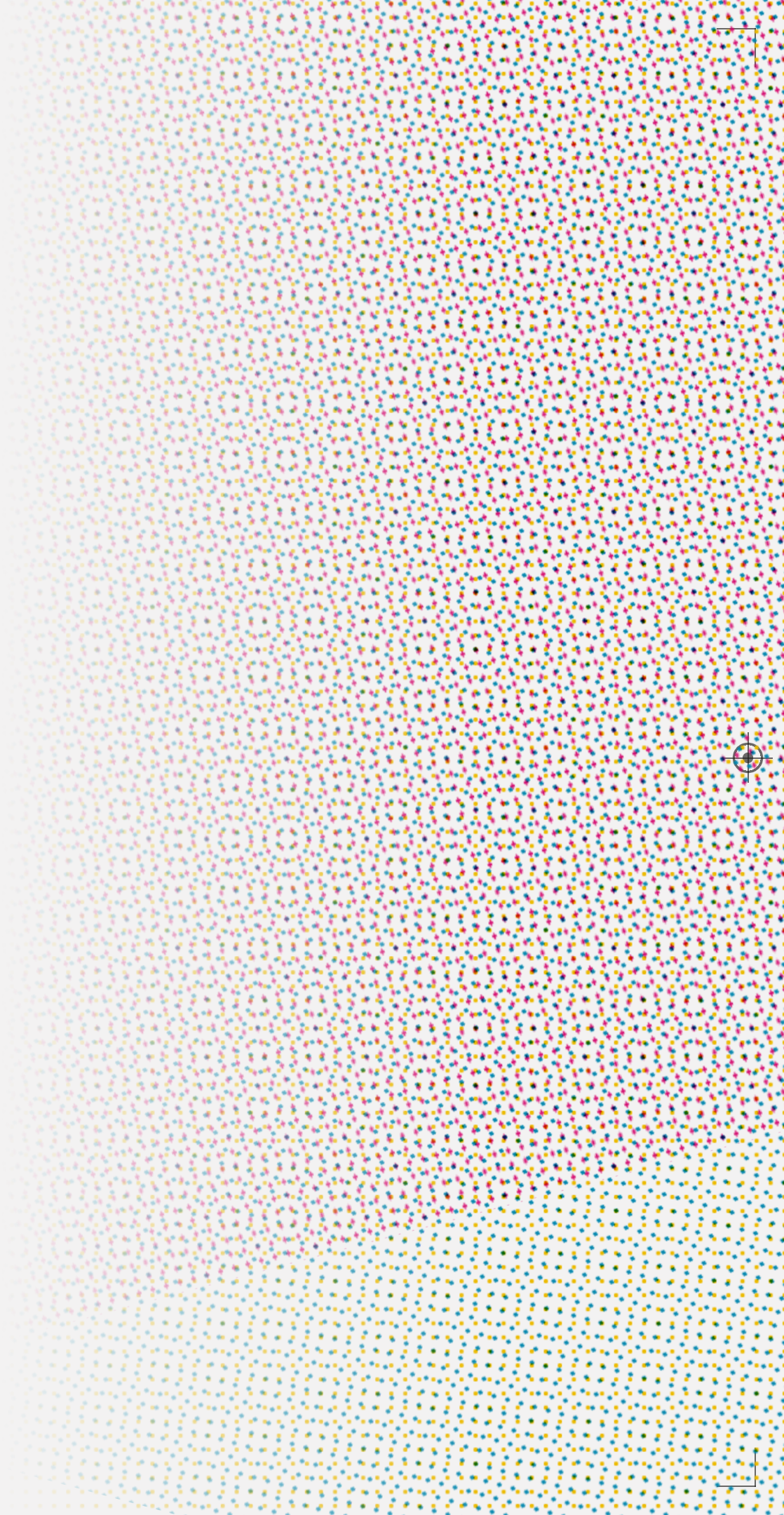
# Five outcome labels

| LABEL | MEANING                            |
|-------|------------------------------------|
| 0     | Candidate declined                 |
| 1     | Recruiter declined the candidate   |
| 2     | Candidate agreed                   |
| 3     | Next steps were discussed          |
| 4     | Candidate actually started the job |

Calls that failed for technical reasons carry no signal about interview outcome and are excluded from training.



# Modeling approach



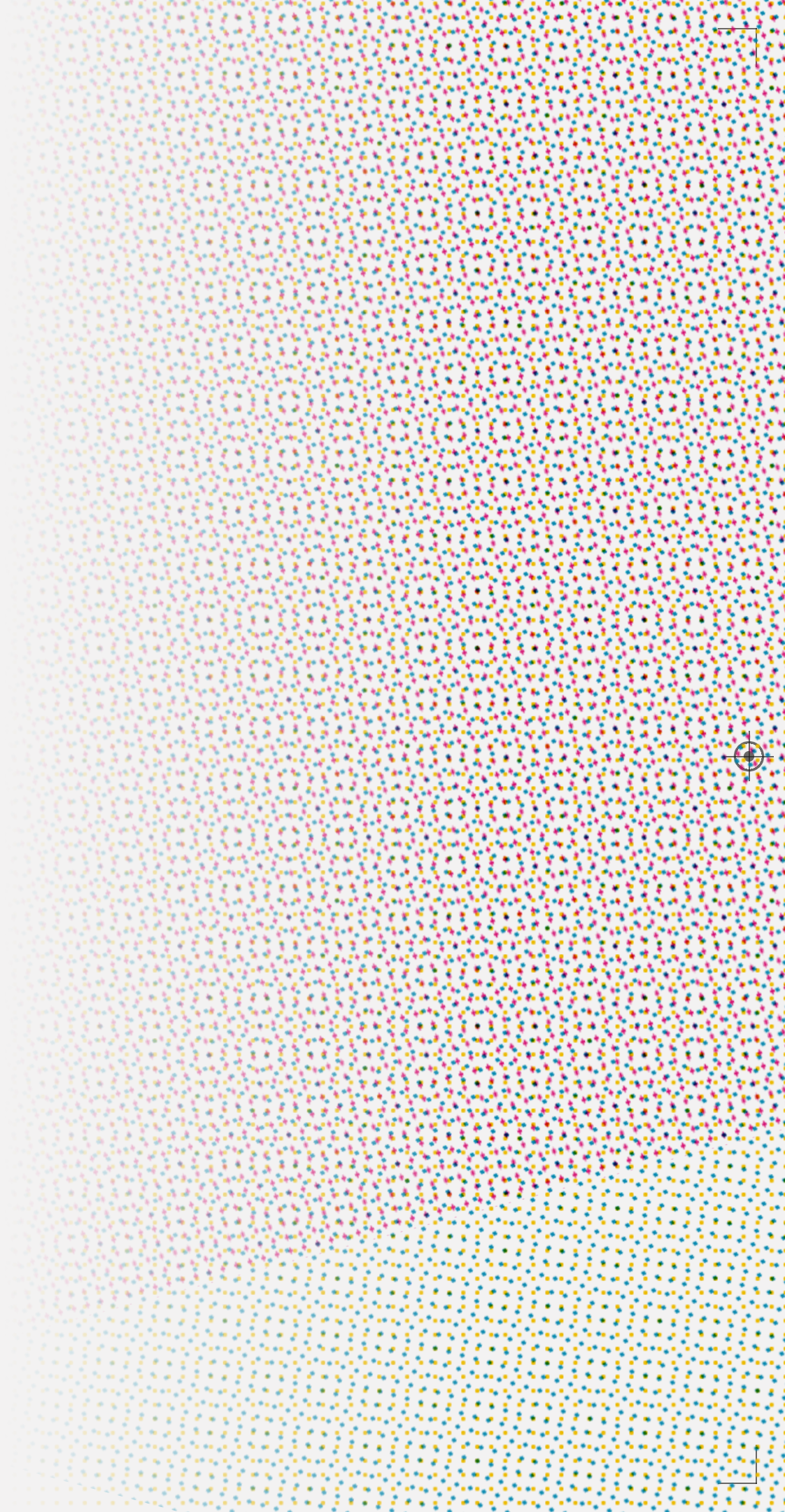
# Fine-tuning ruBERT-base

| PARAMETER         | VALUE                                                     |
|-------------------|-----------------------------------------------------------|
| Base model        | ai-forever/ruBert-base                                    |
| Fine-tuning head  | BertForSequenceClassification — dropout + linear on [CLS] |
| Tokenisation      | WordPiece, max length 512, dynamic padding                |
| Optimiser         | AdamW, lr = 1e-6                                          |
| Batch size        | 4                                                         |
| Epochs            | 3                                                         |
| Train / val split | 80 / 20, stratified by label                              |



# Results

03







# 870%

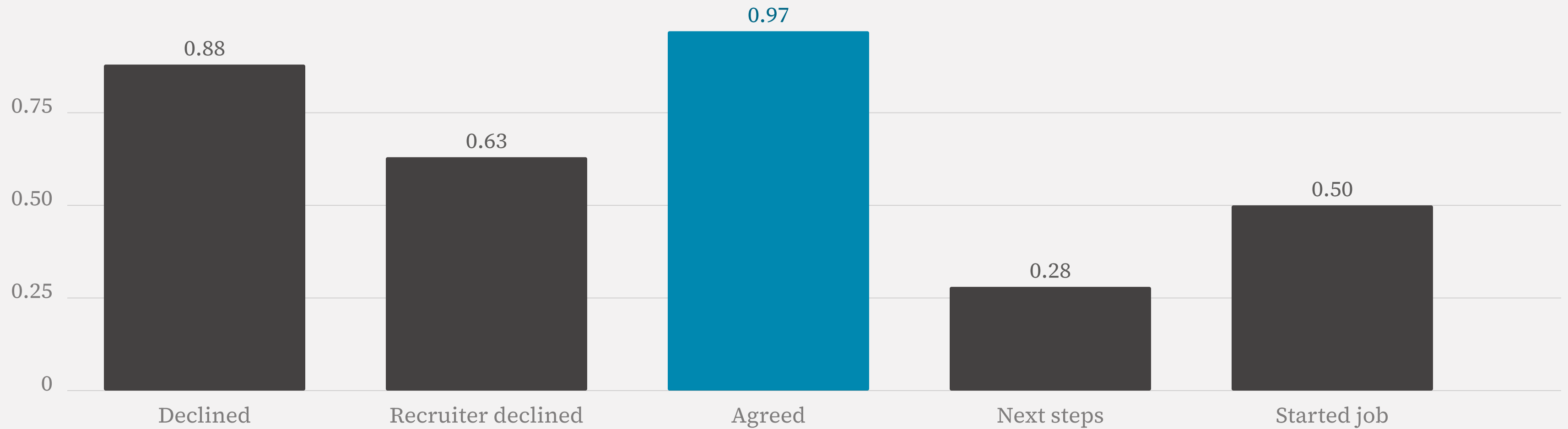
Overall accuracy on a held-out validation set of 2,129 labelled calls — weighted F1 of 0.86 across the five outcome classes.

# Performance by class

| CLASS                        | PRECISION | RECALL | F1   | SUPPORT |
|------------------------------|-----------|--------|------|---------|
| Candidate declined           | 0.86      | 0.90   | 0.88 | 467     |
| Recruiter declined candidate | 0.86      | 0.49   | 0.63 | 77      |
| Candidate agreed             | 0.96      | 0.98   | 0.97 | 1,313   |
| Next steps discussed         | 0.35      | 0.23   | 0.28 | 124     |
| Candidate started job        | 0.47      | 0.54   | 0.50 | 148     |

Macro average F1 0.65, weighted average F1 0.86 — the imbalance between classes drives the gap between the two.

# F1 score by class



Candidate agreed dominates the training data and is classified most reliably; Next steps discussed is the hardest class — it is frequently confused with Candidate started job.



# Confusion matrix

| ACTUAL \ PREDICTED     | 0   | 1  | 2     | 3  | 4  |
|------------------------|-----|----|-------|----|----|
| 0 — Declined           | 420 | 6  | 41    | 0  | 0  |
| 1 — Recruiter declined | 29  | 38 | 10    | 0  | 0  |
| 2 — Agreed             | 24  | 0  | 1,288 | 1  | 0  |
| 3 — Next steps         | 5   | 0  | 2     | 28 | 89 |
| 4 — Started job        | 12  | 0  | 6     | 50 | 80 |

Labels 3 and 4 confuse each other in both directions — both involve future-oriented language, and the boundary between them is semantically thin.

# Limitations & future work

## Class imbalance

Recruiter declined (77 samples) and Next steps discussed (124) are dwarfed by Candidate agreed (1,313). Class-weighted loss or oversampling would likely lift both classes' recall.

## Label ambiguity

Next steps discussed and Candidate started job sit close in meaning. Clearer annotation guidelines, or merging the two classes, would remove much of that confusion.



# Thank you

ruBERT-base fine-tuned for five-way  
interview outcome classification —  
full notebook and README available  
on request.